AI Beauty Contest

LLM-Driven Mission Control for ROS 2 Robotic Inspection

Simulation-First Inspection, Patrolling and Monitoring

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Project website:
apostolosvarelas.github.io/ai_contest_webpage/

Abstract

This report presents a simulation-first mission-control and evidence platform for Unitree Go2 quadruped robots. The system enables an operator to express an inspection, patrolling or monitoring objective in natural language. A repository-local large language model decomposes the objective into a constrained set of robotic tasks, while deterministic planning and safety components validate and execute those tasks through ROS 2.

The prototype integrates natural-language mission planning, guarded navigation, single-robot and four-robot coordination, camera and LiDAR visualization, evidence collection, report generation and deployment history within a unified browser interface. The report describes the initiative and its value, the solution architecture, the implemented robotic skill, the measured key performance indicators and the associated business case. Results are presented as simulation evidence and are not interpreted as proof of physical-robot safety or field performance.

Contents

Chapter 1

Initiative Description and Value

1.1 Problem and Selected Use Case

Facilities and operational sites require recurring inspection, patrolling and monitoring of rooms, work zones, equipment and infrastructure. Suitable indoor settings include warehouses, industrial buildings, utility rooms and other structured facilities, while bounded outdoor settings can include logistics yards, agricultural areas, industrial perimeters and infrastructure sites. These activities are necessary for detecting misplaced assets, obstructions, damaged equipment and other visible anomalies, but they can be repetitive, time-consuming and difficult to document consistently. Some areas may also expose personnel to moving equipment, poor visibility or other operational hazards.

Mobile robots can perform part of this work, but requesting a new mission typically requires knowledge of ROS 2, navigation interfaces, semantic maps, sensor configuration and robot-specific commands. An operational objective such as

> *"Inspect the receiving dock for barrels, save the evidence and produce a findings report."*

may therefore need to be translated manually into routes, observation points, camera actions and reporting steps by a robotics engineer. This creates a gap between the person who understands the operational need and the technical system capable of executing it.

The selected use case is **supervised warehouse inspection**. The warehouse is a concrete implementation example of the broader operational need across the facilities and sites described above, rather than the only intended application. It is a bounded instance of the inspection, patrolling and monitoring applications defined by the challenge. The objective is not to give a language model direct control of a robot. The objective is to allow an operator to express the desired outcome in natural language and have the system convert it into validated, observable and interruptible ROS 2 operations.

The initial customer hypothesis is a warehouse or logistics operator that performs recurring visual inspections and needs a consistent record of the result. The software is intended to complement, rather than replace, existing robotics, safety and operational procedures.

1.2 Target Customer and Operational Need

The initiative distinguishes the user of the application from the people who purchase, approve and integrate it. Their principal needs are summarized in Table 1.1.

The primary operational need can therefore be expressed as follows:

> **Enable a non-robotics operator to request, supervise and document a bounded inspection of a configured operational site while deterministic software retains authority over physical execution and safety-related decisions.**

This need is commercially relevant only if the resulting workflow proves more usable, repeatable or economical than the customer's existing process. The current prototype demonstrates the technical workflow in simulation; customer baseline measurements and physical validation remain part of the proposed pilot stage.

Table 1.1: Target stakeholders and the needs addressed by the initiative.

Stakeholder	Operational need	Application response
Operations, maintenance or safety operator	Request and supervise an inspection without constructing ROS 2 commands or robot-specific routes.	Natural-language mission requests, visible task progress, camera and LiDAR views, evidence review and independent human controls.
Warehouse or facility manager	Obtain repeatable inspection coverage and evidence associated with a known mission, location and time.	Validated semantic tasks, configured viewpoints, mission history and evidence-based findings reports.
Automation or robotics lead	Introduce new missions without allowing generative output to bypass navigation and safety constraints.	Strict task schemas, deterministic route planning, bounded execution and explicit readiness checks.
Robotics systems integrator	Configure the solution for another site, robot profile or inspection workflow without rewriting the complete application.	Configuration-driven maps, rooms, inspection viewpoints, recognition targets, robot profiles and launch parameters.

1.3 Proposed Initiative

The initiative is an LLM-driven mission-control and evidence platform for Unitree Go2 quadruped robots, implemented as two ROS 2 packages and validated first in MuJoCo simulation. It combines the following capabilities:

- natural-language interpretation using a repository-local Qwen3-4B model executed through llama.cpp;
- deterministic fallback handling for supported instructions;
- a strict mission schema containing predefined navigation, patrol, capture, inspection, return and reporting tasks;
- semantic resolution of named rooms and operational areas;
- deterministic route planning and guarded goal execution;
- independent Stop, cancellation, manual takeover and recovery paths;
- camera and LiDAR visualization through a browser interface;
- evidence collection with mission, robot, task, location, viewpoint, timestamp and pose identity;
- findings-report generation when explicitly requested;
- single-robot operation and scenario-bounded coordination of four simulated robots.

The language model interprets the operator's objective and proposes semantic work. It cannot publish actuator commands, invent physical coordinates, generate executable code, choose arbitrary filesystem paths or bypass configured mission constraints. Deterministic components validate the proposal, resolve the physical details and decide whether execution is permitted.

During execution, the operator supervises the mission through a unified browser workspace containing the semantic map, three-dimensional scene, live camera feeds, LiDAR visualization, task state, mission progress and collected evidence. The application therefore covers the complete simulated workflow from operational request to documented result.

The principal contribution is therefore not the language model in isolation. It is the controlled integration of natural-language interpretation, deterministic mission planning, guarded ROS 2 execution, human authority and evidence management within one observable workflow.

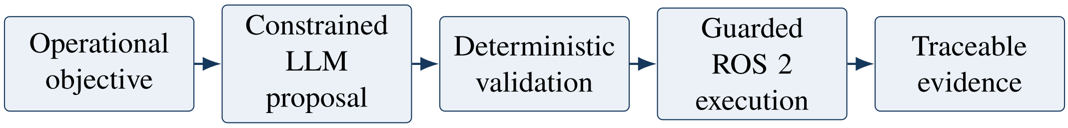

Figure 1.1: Transformation of an operational objective into a bounded, executable and traceable robotic mission.

1.4 Customer Value and Differentiation

1.4.1 Value Proposition

The proposed value is a reduction in the technical effort required to request, supervise and document a robotic inspection. Operators communicate at the level of operational objectives, rooms and assets, while the platform converts those objectives into validated robot tasks.

The resulting value proposition is:

> **Turn an inspection objective for a configured operational site, expressed in natural language, into a guarded and traceable ROS 2 mission, without requiring the operator to write robot commands and without granting the language model physical control authority.**

The initiative is expected to create value in four areas:

- **Accessibility:** operational personnel can request supported missions without constructing ROS 2 command sequences;
- **Repeatability:** inspections use configured routes, viewpoints and task contracts rather than an informal sequence of actions;
- **Human authority:** Stop, cancellation, manual takeover and recovery remain independent of the language model;
- **Traceability:** evidence is associated with the mission, robot, task, location, viewpoint, timestamp and measured pose that produced it.

1.4.2 Differentiation

Table 1.2 distinguishes the proposed initiative from representative alternatives.

Repository-local language and vision models also reduce dependence on public inference services. This supports future on-premises evaluation where facility imagery, maps and operational instructions should remain within the deployment environment. This is a deployment characteristic of the prototype, not yet a validated customer purchasing requirement.

1.5 Value Measurement

The customer value has not yet been measured at a physical site. To avoid presenting hypotheses as established outcomes, the initiative defines how each proposed benefit will be tested.

The corresponding measures and required comparisons are summarized in Table 1.3.

The challenge results for mission completion, human intervention and visual classification are reported in Chapter 5. Customer labour savings, return on investment and physical-site performance remain future pilot measurements.

Table 1.2: Differentiation from representative inspection approaches.

Approach	Typical limitation	Proposed differentiation
Manual inspection	Requires a person to perform and document every patrol directly.	Supports repeatable simulated robot missions with automatically associated evidence while retaining human supervision.
Fixed scripted robot mission	A changed objective may require an engineer to edit or trigger a different predefined sequence.	Natural-language interpretation can compose supported semantic tasks without changing the physical execution interfaces.
Engineering-led ROS 2 control	Exposes technically powerful interfaces but requires robotics knowledge from the mission author.	Presents rooms, assets and operational objectives through an operator-facing mission workspace.
Unrestricted LLM-to-robot control	Generative output may influence physical execution without a sufficiently narrow authority boundary.	Restricts the model to semantic proposals and keeps coordinates, routing, safety checks and execution deterministic.
Disconnected evidence workflow	Images and findings may require manual association with the mission that produced them.	Preserves evidence provenance across runtime, location, mission, robot, task, viewpoint, timestamp and pose.

Table 1.3: Value hypotheses and their corresponding evaluation measures.

Value hypothesis	Evaluation measure	Required comparison
Easier mission creation	Time and operator actions required to request an accepted mission.	Natural-language workflow compared with the existing engineering-led procedure.
Reliable mission execution	Mission completion rate and bounded failure reasons.	Repeated execution of a predefined mission suite under documented conditions.
Reduced supervision burden	Human intervention rate and intervention type.	Number of interventions divided by attempted missions, with manual test interventions reported separately.
Reliable visible findings	False-positive and false-negative rates for the selected targets.	Qualified perception output compared with labelled ground truth.
Complete inspection evidence	Proportion of required viewpoints with correctly attributed accepted evidence.	Expected viewpoint set compared with stored evidence records.
Faster documentation	Time from mission completion to an available findings report.	Proposed evidence workflow compared with the current reporting procedure.

1.6 Scalability and Reusability

The platform is designed around a reusable mission-control core and configuration-driven deployment data. The reusable core contains the LLM interface, task schema, mission state model, task compiler, routing and safety logic, ROS 2 integration, operator interface, perception workflow and evidence pipeline.

Site-specific configuration defines:

- semantic maps and traversable regions;
- named rooms, zones and aliases;
- patrol seeds and inspection viewpoints;
- recognition targets and mission presets;
- scene assets and terrain information;
- robot, sensor and locomotion parameters.

This separation is important because a scalable solution cannot require the entire mission-control application to be rewritten for every site. New environments still require mapping, inspection design and safety validation, but they can reuse the same mission vocabulary and execution pipeline.

Table 1.4 summarizes the implemented reuse mechanisms and their current evidence boundaries.

Table 1.4: Implemented scalability mechanisms and their current evidence.

Dimension	**Implemented mechanism**	**Current evidence and boundary**
Mission reuse	Shared semantic task schema and composite inspection workflow.	The same task concepts support navigation, patrol, capture, inspection, return and reporting requests.
Environment reuse	Configuration-driven maps, rooms, routes, viewpoints and targets.	Demonstrated across warehouse, agricultural and scanned real-location scenarios in the repository.
Robot configuration	Robot and locomotion profiles are resolved independently from the natural-language task model.	Multiple repository-bundled locomotion profiles are selectable, but physical robot portability is not yet demonstrated.
Fleet extension	Robot-specific namespaces, task allocation, room ownership and route coordination.	Demonstrated with four robots in one bounded MuJoCo warehouse scenario; production fleet scale is not established.
Model deployment	Repository-local, hash-verified language and vision assets.	Supports repeatable local setup while remaining dependent on suitable host compute and the documented software environment.

The repository therefore demonstrates **software scalability through reuse and configuration**, not production-scale autonomous deployment. The next validation step is to measure how much time and customer-specific engineering are required to configure an additional site and robot.

Chapter 2

Solution Architecture

2.1 Architectural Principle

The architecture translates a natural-language objective into a bounded and observable ROS 2 mission. Its central design principle is the separation of **semantic interpretation** from **physical execution**. The language model may propose what the robot should do, but deterministic components decide whether the proposal is valid and how it can be executed.

This separation addresses four requirements of the challenge:

- the operator can describe inspection, patrolling and monitoring objectives in natural language;
- the language model decomposes the objective using only predefined robotic tasks;
- semantic tasks are validated before they reach ROS 2 execution;
- stopping, manual intervention and safety decisions remain independent of generative inference.

The resulting architecture is deliberately restrictive. The language model cannot publish actuator commands, generate executable code, select arbitrary filesystem paths or bypass the configured map and safety constraints.

2.2 System Overview

The application is organized into presentation, interpretation, mission management, safety, execution and evidence layers. Figure 2.1 shows their logical relationships.

Browser communication uses local HTTP and WebSocket interfaces. Communication between the mission, perception and simulation processes uses ROS 2 topics and services. The portal can start one selected simulation runtime on demand and reuse the verified local language-model service across location changes. The fleet scenario instead starts four namespaced robot controllers in one shared MuJoCo world.

2.3 Component Responsibilities and Traceability

Table 2.1 connects each architectural layer to its responsibility and principal implementation. The listed paths are relative to the two packages under `src/`.

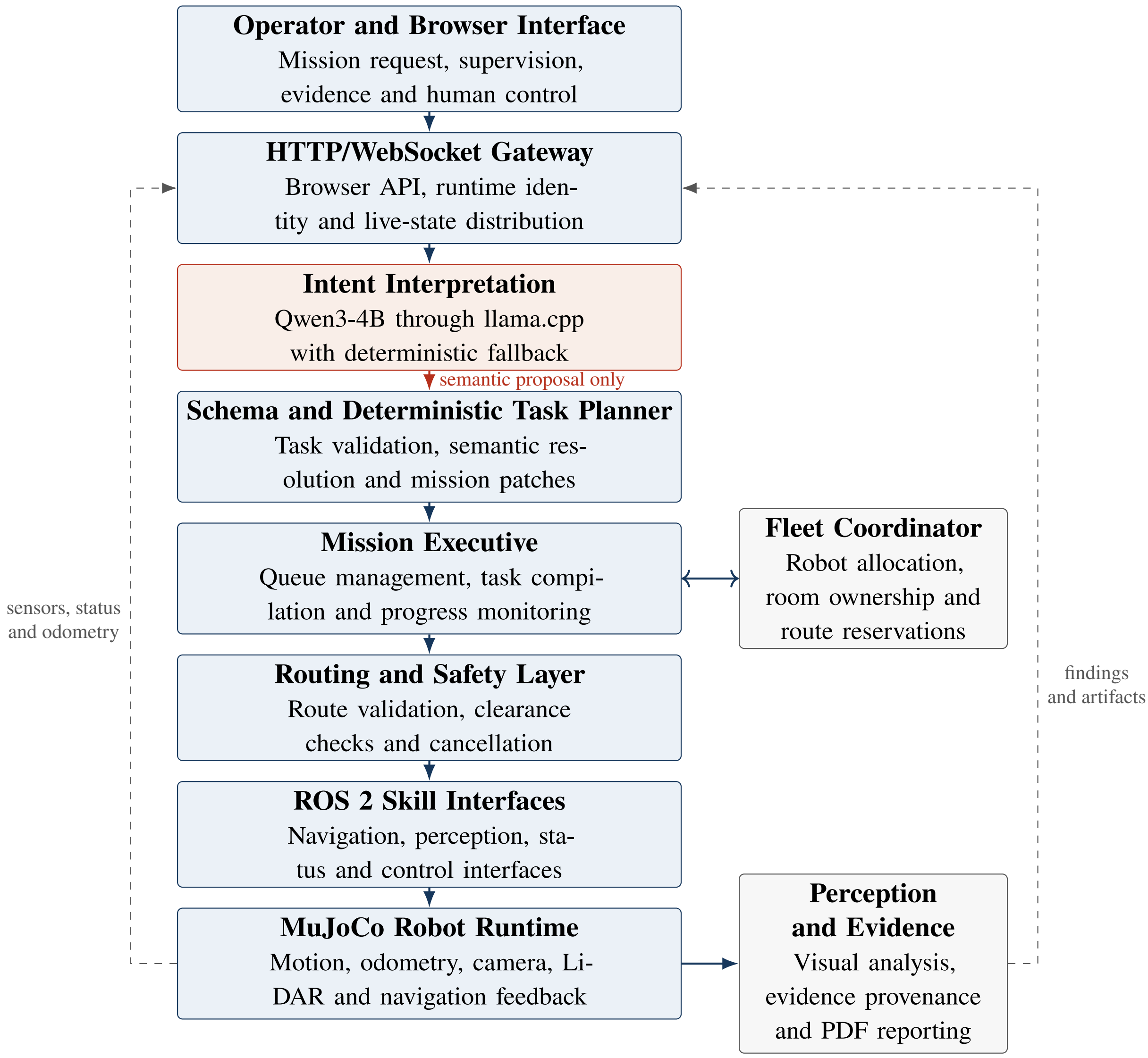

Figure 2.1: Logical architecture and authority boundary of the mission-control platform. The language model has no direct execution path to the robot.

Table 2.1: Responsibilities and implementation traceability of the principal architectural layers.

Layer	Responsibility	Principal implementation
Presentation and gateway	Accepts operator requests and presents mission state, cameras, LiDAR, evidence and independent human controls.	`web/app.py`, `web_gateway_node.py`
Intent interpretation	Converts natural language into a semantic mission proposal using the local model or bounded deterministic handling.	`llm_planner.py`, `intent_parser.py`
Schema and mission model	Restricts task types and fields, validates values and applies revision-checked mission modifications.	`schema.py`, `mission.py`
Executive and skill compiler	Converts accepted tasks into finite execution steps, manages their lifecycle and records outcomes.	`task_execution.py`, `executive_node.py`
Routing and safety	Resolves valid routes, checks clearance and prevents stale, blocked or unsupported goals from reaching execution.	`planner.py`, `route_alternatives.py`, `navigation_safety.py`
Fleet coordination	Allocates robots and coordinates rooms, routes, peers and charging resources without delegating arbitration to the language model.	`fleet_coordinator_node.py`, `fleet_scheduler.py`
Perception and evidence	Processes authorized visual requests and preserves evidence identity, qualification and report provenance.	`perception_worker_node.py`, `perception_validation.py`, `reporting.py`
Robot runtime	Simulates locomotion and publishes odometry, camera, LiDAR and navigation status.	`simulator_node.py`, `fleet_simulator_node.py`

2.4 Mission Processing and Authority

A natural-language request is processed as a sequence of constrained transformations:

1. The browser submits the objective together with the active runtime and mission context.
2. Qwen3-4B, or the deterministic fallback, proposes a mission using the supported semantic task vocabulary.
3. The schema rejects unknown fields, invalid values, unsupported tasks and unresolved semantic locations.
4. A revision check prevents a delayed planning response from overwriting a newer operator instruction.
5. The mission executive compiles each accepted task into finite navigation, capture, inspection or reporting steps.
6. Deterministic routing and safety components validate physical goals before publishing them through ROS 2.
7. Runtime feedback updates task progress, while accepted evidence is associated with the active robot, mission, task, location, viewpoint, timestamp and pose.

The model may propose the following task categories:

- `navigate_room` and `navigate_point`;
- `patrol_room`;
- `capture_room` and `inspect_room`;
- `return_station`;

- `report`.

Named locations are converted into physical coordinates by deterministic map logic rather than by the language model. Likewise, route selection, fleet allocation, perception qualification and report authorization remain outside generative inference.

Stop, cancellation, manual takeover and upright recovery use separate control paths. They can therefore pre-empt an active mission without waiting for the language model or accepting a new plan.

2.5 Principal ROS 2 Interfaces

Table 2.2 lists the interfaces most relevant to the end-to-end mission path. JSON-structured application state is transported in bounded `std_msgs/msg/String` messages, while standard ROS 2 types are used for physical goals and sensor data.

Table 2.2: Principal ROS 2 interfaces in the mission execution path.

Interface	Type	Purpose
`/go2/demo/command`	`std_msgs/String`	Transfers the operator request from the gateway to the mission executive.
`/go2/demo/state`	`std_msgs/String`	Publishes the authoritative mission and task state.
`/go2/nav/goal` and `/go2/nav/status`	`PoseStamped`; `String`	Transfer validated navigation goals and return navigation status.
`/go2/demo/perception/request` and `/go2/demo/perception/result`	`std_msgs/String`	Exchange bounded, identity-checked perception work and results.
`/go2/odom`, `/go2/front_camera/image_raw` and `/go2/lidar/points`	`Odometry`; `Image`; `PointCloud2`	Provide robot pose, visual observations and range measurements.
`/go2/demo/stop`, `/go2/nav/cancel` and `/go2/recover_upright`	`Trigger` services	Provide independent stopping, navigation cancellation and recovery.
`/go2/demo/manual_override` and `/go2/manual/cmd_vel`	`SetBool`; `Twist`	Transfer and release control authority for bounded manual motion.

Fleet operation applies the same interface roles under robot-specific namespaces. This preserves per-robot state while allowing the fleet coordinator to maintain shared room and route constraints.

2.6 Safety and Failure Handling

The architecture fails closed when a mandatory precondition is not satisfied. Table 2.3 summarizes representative failure paths.

Additional safeguards include bounded task and inference queues, finite retry budgets, evidence-identity validation and explicit authorization for persistent media or report generation. These controls constrain the demonstrated simulation behaviour; they do not constitute certified functional safety for a physical robot.

Table 2.3: Representative failure conditions and architectural responses.

Condition	Deterministic response	Result
Invalid model output	Reject unknown tasks, fields or out-of-range values.	No robot command is issued.
Ambiguous or unknown location	Require exact semantic-map resolution.	Clarification or bounded failure.
Stale plan or runtime identity	Compare mission revision and runtime generation.	Outdated work is rejected or quarantined.
Blocked or unsafe goal	Reject goals outside traversable space or without sufficient route clearance.	Mission remains stopped or attempts an eligible alternative.
Missing model or sensor	Apply readiness, freshness and timeout checks.	Dependent functionality remains unavailable.
Operator intervention	Invoke Stop, cancellation or manual ownership through the independent control path.	Active navigation is cancelled or held.
Fleet conflict	Enforce room ownership, route admission and peer-aware coordination.	Conflicting work waits or is reassigned.

Chapter 3

Documented and Working Robotic Skill

3.1 Skill Definition and Purpose

The principal skill documented for the challenge is `inspect_room`. It implements a bounded asset-inspection workflow that combines semantic mission interpretation, navigation, visual observation and evidence collection.

Its purpose is defined as follows:

> Navigate through the configured inspection viewpoints of a semantic area, collect the requested visual observations and retain traceable evidence without allowing the language model to generate physical control commands.

The skill is composite rather than a monolithic ROS 2 node. Its behaviour is distributed across the task schema, semantic map, task compiler, mission executive, route planner, simulator and perception worker. This separation allows each stage to be validated independently.

The skill does not generate a findings report automatically. Reporting is a separate `report` task and requires an explicit operator request. This prevents an inspection request from implicitly authorizing a persistent filesystem export.

3.2 Skill Contract

Natural-language input is converted into a validated `TaskSpec`. The accepted fields are summarized in Table 3.1.

The skill returns:

- a task status of `queued`, `running`, `cancelling`, `succeeded`, `failed` or `cancelled`;
- the compiled route and available planning diagnostics;

Table 3.1: Input contract of the configured-area inspection skill.

Field	Type	Constraint
`kind`	Constant	Must be `inspect_room`.
`room_id`	String	Must resolve to exactly one room in the active semantic map.
`queries`	List of strings	Optional visual targets such as `barrel`. At most 32 queries are accepted, with a maximum length of 160 characters per query.
`capture_interval_s`	Number	Accepted range of 0.2 s to 60 s; the default is 1 s.
`max_frames`	Integer	Accepted range of 1 to 256. This bound does not remove mandatory spatial coverage.
`save_images`	Boolean	Enables persistent image retention only when requested; the default is `false`.

- structured observations for the authorized visual queries;
- evidence associated with the mission, robot, task, room, viewpoint, timestamp and measured pose;
- a completion summary or bounded failure reason.

Task success and perception success are intentionally different concepts. A task may complete every required navigation and capture step while producing an inconclusive visual observation. Conversely, an object is reported as a confirmed finding only when the active perception qualification permits that claim.

3.3 Preconditions

The skill can enter execution only when an active runtime and validated semantic map exist for the selected location. The requested room must resolve unambiguously and provide at least one usable inspection pose or validated patrol seed, while current robot odometry must be available and belong to the active runtime generation.

Every mandatory inspection viewpoint must have a traversable route. When the request includes visual analysis, the front-camera and perception channels must also be ready. Manual control or recovery must not own the robot; in fleet mode, the coordinator must additionally have granted the required room and route admission.

If a mandatory precondition is not satisfied, physical execution is not dispatched. The task instead produces a clarification or bounded failure describing the rejected condition.

3.4 Execution Procedure

The accepted semantic task is first validated against the supported fields, value ranges, capture bounds and semantic references. The deterministic map logic then loads the requested room and its configured inspection viewpoints; the language model cannot invent viewpoints or physical coordinates. A collision-aware ordering covers every required viewpoint and is compiled into bounded navigation segments.

For each segment, the executive publishes the validated goal on `/go2/nav/goal` and monitors odometry, navigation status, obstruction, timeout and cancellation. At the required pose and heading, it publishes an identity-bound perception request for the authorized queries. The resulting observation is accepted only when its runtime, location, robot, task, room, camera, viewpoint and pose identities match the active request.

Execution repeats until every mandatory viewpoint has been processed. The executive then clears the active goal and temporary execution state before publishing the terminal result.

This procedure ensures that an operator-specified image count cannot replace mandatory viewpoint coverage. It also prevents evidence from an earlier mission, robot or runtime from being attributed to the current inspection.

3.5 Skill State Model

Figure 3.1 presents the principal execution states. Stop or cancellation can pre-empt navigation and perception without waiting for language-model inference.

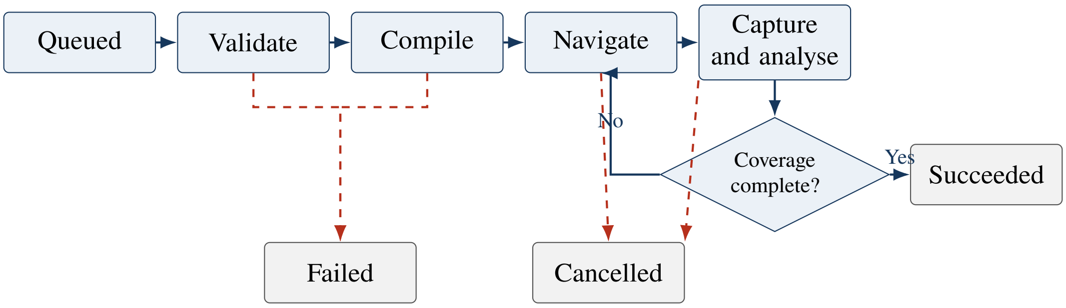

Figure 3.1: Simplified state model of the `inspect_room` skill. Dashed transitions represent rejection, failure or operator cancellation.

An unrecoverable failure during navigation, capture or perception also leads to `failed`. Temporary obstructions may instead invoke bounded replanning when an eligible alternative route exists.

3.6 Postconditions

Table 3.2 defines the terminal contract.

Table 3.2: Postconditions of the inspection skill.

Terminal state	**Postconditions**
`succeeded`	Every mandatory viewpoint has been processed; required headings and captures have completed; accepted evidence has the correct provenance; active goals and temporary execution state have been cleared.
`failed`	Execution has stopped; no additional views are requested; the failure reason is recorded; partial evidence remains explicitly incomplete and retains only its actual provenance.
`cancelled`	Active navigation has been cancelled or brought to a controlled hold; pending perception work is cancelled or quarantined; fleet resources are released when safe to do so.

No terminal state authorizes a findings PDF implicitly. A PDF is produced only if a separate, explicit `report` task satisfies its own authorization and evidence requirements.

3.7 Safety Constraints

The explicit safety constraints of the skill are listed in Table 3.3.

Table 3.3: Safety constraints of the `inspect_room` skill.

Constraint	**Implemented behaviour**
Constrained model authority	The language model selects semantic task fields only. It cannot generate actuator commands, executable code, inspection geometry or ROS 2 interfaces.
Semantic and route validity	Unknown rooms, goals outside traversable space and disconnected routes are rejected rather than approximated.
Complete spatial coverage	Capture preferences cannot remove mandatory inspection viewpoints.
Bounded execution	Query counts, frame counts, queues, retries and timeouts have explicit upper bounds.
Runtime and evidence identity	Stale commands or observations from another runtime, location, mission, robot or task are rejected or quarantined.
Human authority	Stop, cancellation, manual takeover and recovery can pre-empt the skill without consulting the language model.
Fleet admission	Room ownership and route coordination prevent conflicting fleet work from being dispatched simultaneously.
Evidence qualification	Unqualified, stale or spatially inconsistent observations cannot be presented as confirmed findings.
Explicit persistence	Permanent images and findings reports require explicit operator authorization.

These mechanisms constrain behaviour within the simulation. They do not replace a hardware emergency stop, certified functional-safety system or site-specific risk assessment for a physical robot.

3.8 Implementation and Verification

The skill is traceable from its external contract to production code and automated verification, as shown in Table 3.4. Production paths are relative to `src/go2_ai_demo/go2_ai_demo/`; test paths are relative to `src/go2_ai_demo/test/`.

Table 3.4: Implementation and automated verification of the inspection skill.

Concern	**Production implementation**	**Automated evidence**
Contract and mission state	`schema.py`, `mission.py`	`test_schema_and_mission.py`
Viewpoint coverage and compilation	`task_execution.py`, `semantic_map.py`	`test_task_execution.py`
Execution and goal cleanup	`executive_node.py`	`test_navigation_stop_goal_cleanup.py`, `test_task_execution.py`
Perception qualification	`perception_worker_node.py`, `perception_validation.py`	`test_perception_validation.py`, `test_perception_verifier.py`
Evidence identity	`perception.py`, `web_gateway_node.py`	`test_evidence_identity_quarantine.py`
Cancellation and human control	`executive_node.py`, `web_gateway_node.py`	`test_runtime_stop.py`, `test_manual_control_gateway.py`
Evidence-based reporting	`reporting.py`	`test_reporting.py`, `test_report_progress.py`

The automated tests provide repeatable verification of individual contracts and failure paths. The end-to-end demonstration in Chapter 4 provides the complementary integration evidence, while aggregate mission and perception results are reported in Chapter 5.

3.9 Representative Execution

A representative operator instruction is:

> *"Inspect the receiving dock for barrels and save the inspection images."*

After semantic resolution, the validated task is equivalent to:

```
{
  "kind": "inspect_room",
  "room_id": "receiving_dock",
  "queries": ["barrel"],
  "capture_interval_s": 1.0,
  "max_frames": 12,
  "save_images": true
}
```

The resulting execution trace is:

```
request -> schema validation -> room resolution
        -> viewpoint and route compilation
        -> navigate -> capture -> validate evidence
        -> repeat until coverage is complete
        -> publish terminal task status
```

A successful execution confirms that the required movement and observation procedure completed. The presence of a barrel is reported separately as confirmed, absent, candidate or inconclusive according to the active perception qualification. This distinction prevents successful navigation from being misrepresented as successful object recognition.

Chapter 4

End-to-End Demonstration

4.1 Choose an Operation

This chapter is a self-contained visual walkthrough of the running platform. The figures show what an operator sees and how a natural-language request moves from mission creation to robot execution, evidence review and reporting. No external media is required to understand the workflow.

After entering mission control, the operator first chooses the operational family. The default mission-library view presents two explicit choices: **Patrolling** for monitoring and inspection, and **Rescuing** for search and response. Each card explains its purpose, reports the number of configured locations and opens the corresponding deployment map. This prevents the operator from mixing site presets and workflows from different operational contexts.

Moving over Patrolling gives immediate visual confirmation of the selected monitoring and inspection context before the operator opens its location map. The card describes routine warehouse and agricultural patrols, route monitoring, inspections and findings management.

Moving over Rescuing previews the search-and-response context. Its description sets a different operational expectation: remote search support, safer access and live evidence collection in complex industrial and mine environments.

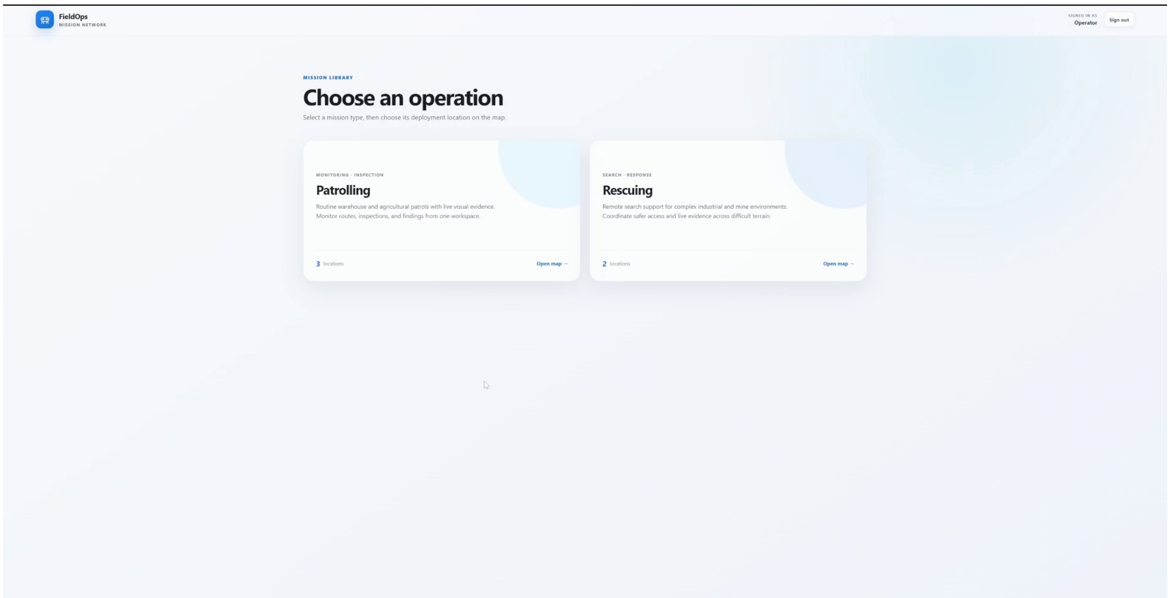

Figure 4.1: Default operation-selection screen. Patrolling and Rescuing are presented as separate mission families, with a short operational description, configured-location count and map entry point.

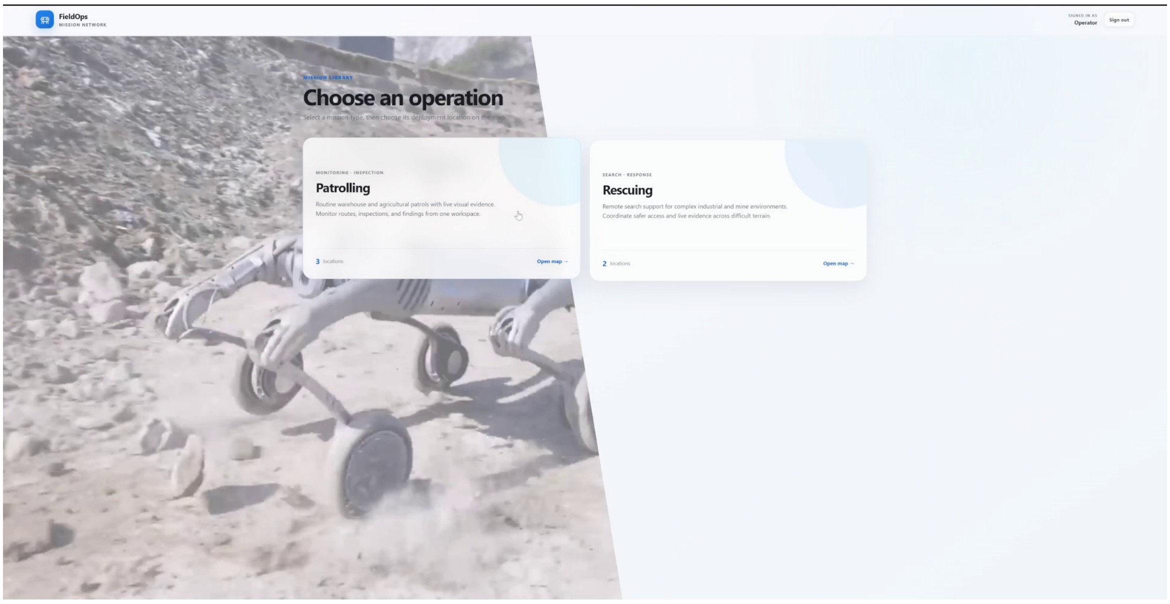

Figure 4.2: Patrolling operation preview. The background changes to the patrolling context while the card keeps its supported workflow and three configured locations visible.

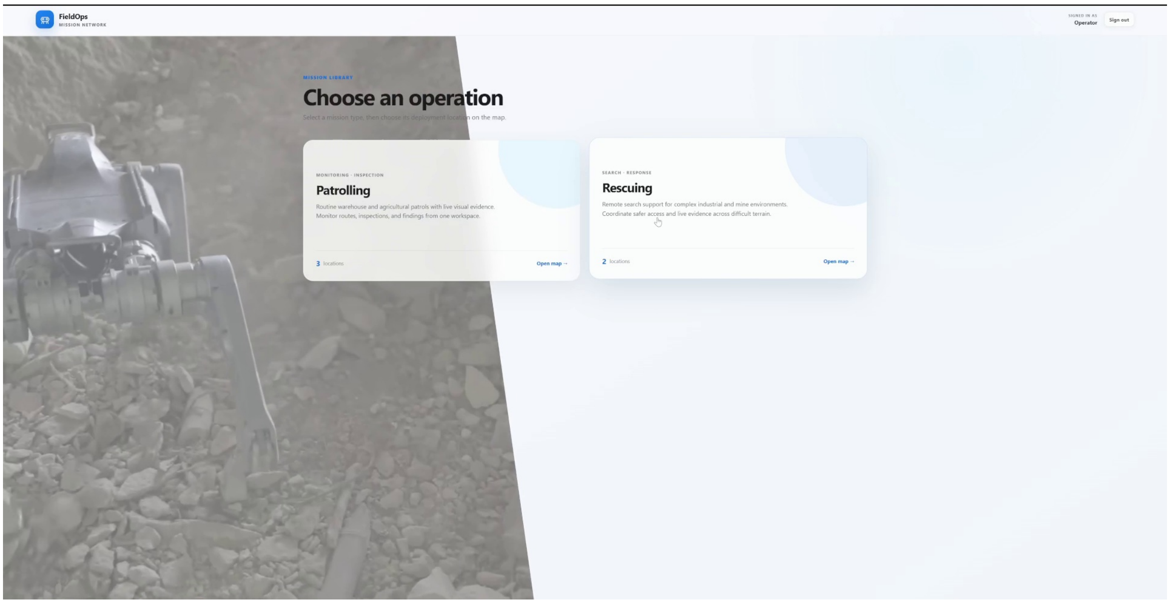

Figure 4.3: Rescuing operation preview. The search-and-response context is visually selected while the card identifies its two configured deployment locations and dedicated map entry point.

4.2 Platform Workspace

The principal workspace is intentionally organized around one supervisory screen. The left column contains the mission conversation, live activity cards, the task board, evidence folders and the instruction box. The right side contains the spatial view, live LiDAR and robot camera. Navigation, 3D, plan and camera modes change the main view without hiding mission state, while the Stop control remains available during execution.

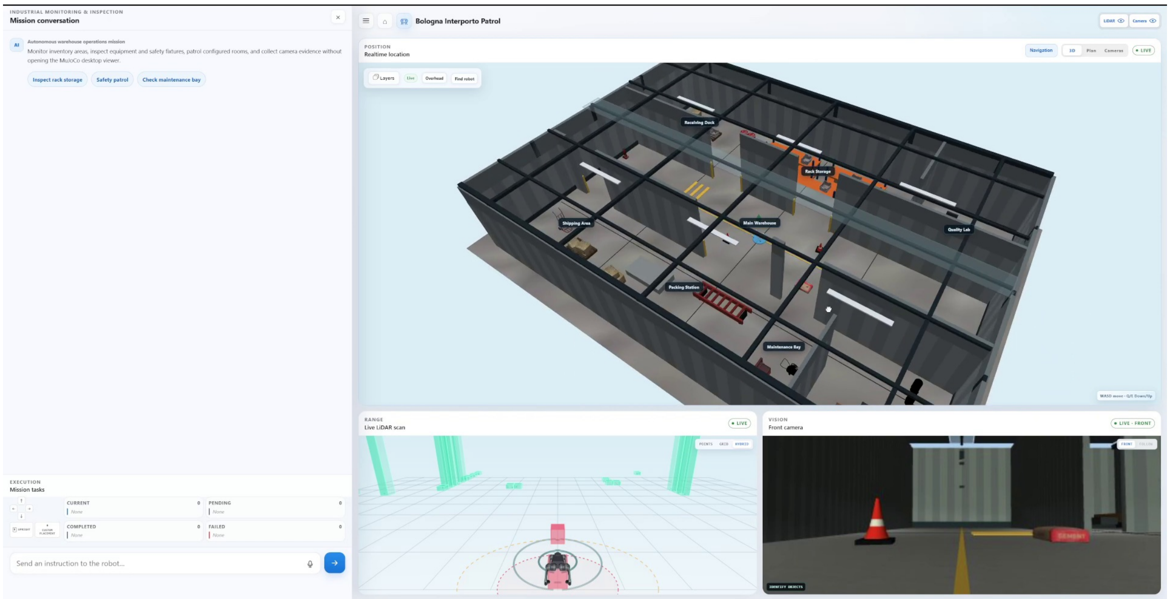

Figure 4.4: Single-robot operator workspace. Suggested missions and free-text input are on the left; the configured semantic environment, LiDAR and front camera are on the right; mission state and collected evidence remain visible below the conversation.

4.3 Create a Mission in Natural Language

The operator does not need to write ROS 2 commands or assemble a task graph. For the representative receiving-dock workflow, the operator enters:

> *"Go to the receiving dock, do an inspection and report it."*

The assistant resolves the configured location and expands the objective into three bounded tasks: navigate to the receiving dock, inspect the receiving dock and produce a findings report. The resulting queue is exposed to the operator; navigation is current, inspection and reporting are pending, and the planned route is drawn in the spatial view. The LLM therefore supplies semantic intent, while the task schema and mission executive determine what can execute.

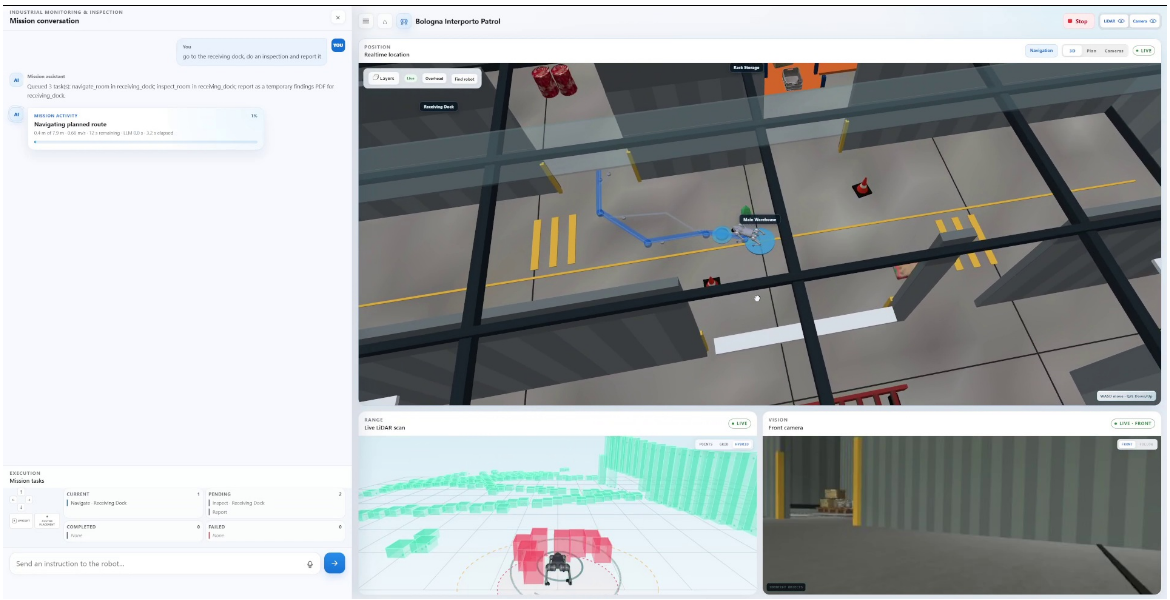

Figure 4.5: Natural-language mission creation. The original sentence, resolved task sequence, active navigation route, task-board state, live LiDAR, front camera and Stop control are visible together.

4.4 Supervise Navigation and Inspection

Once accepted, the mission executive compiles the semantic tasks into guarded robot actions. The operator can follow the robot and route in 3D, switch to the topological plan, inspect the LiDAR safety envelope and observe the front camera. Progress cards distinguish navigation, inspection, analysis and report generation instead of presenting the mission as one opaque activity.

In Figure 4.6, navigation and inspection are complete, report generation is active, and the receiving-dock folder contains the captured observations. The workspace keeps the mission, task, room and evidence context on screen together, which helps the operator understand exactly what is running and where its data came from.

The Stop control is persistent during autonomous execution. It follows a dedicated cancellation path rather than being interpreted by the LLM, so the operator can pre-empt navigation or inspection even if mission reasoning or a perception request is still active.

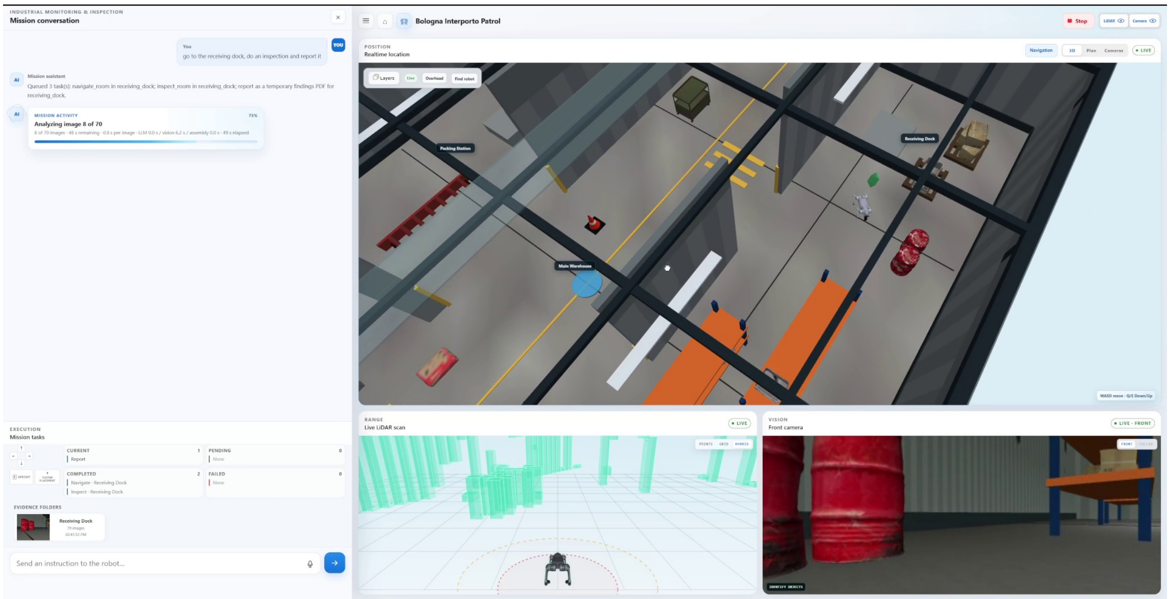

Figure 4.6: Live inspection and evidence processing. Completed and current tasks are explicit; the evidence folder is bound to the receiving dock; and the spatial, LiDAR and camera panes provide simultaneous execution context.

4.5 Review Evidence, Not Only a Chat Answer

Selecting an evidence folder opens the evidence browser. Frames are grouped by task and room, retain capture times, and mark replayed keyframes. Unusable or empty frames are retained as part of the audit trail rather than silently removed; the operator can therefore distinguish what the robot captured from what the perception stack was able to use.

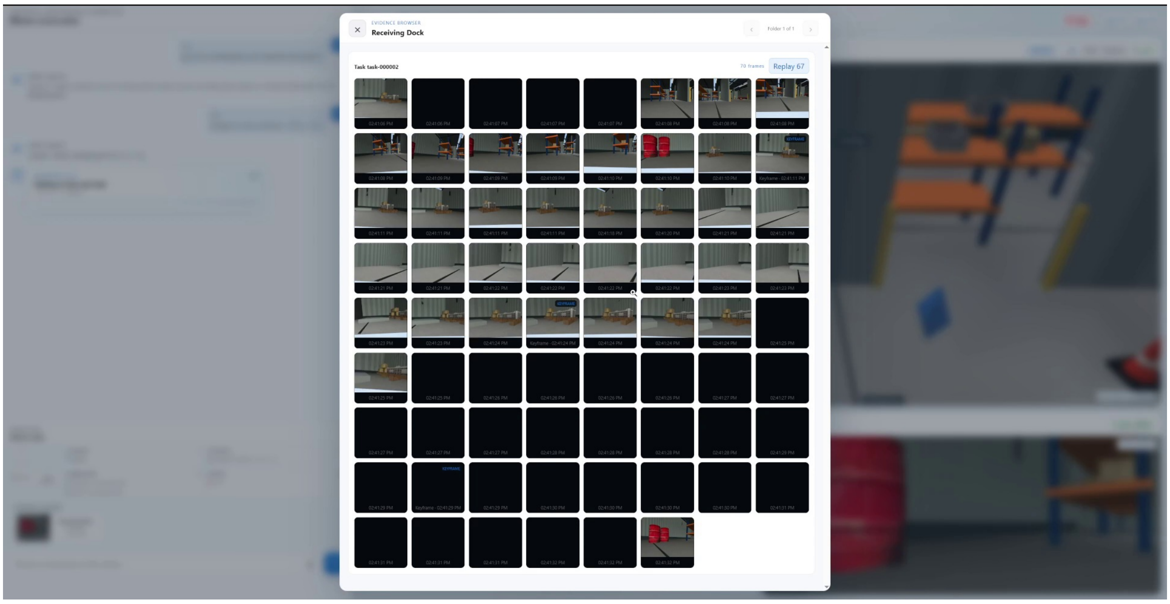

Figure 4.7: Receiving-dock evidence browser. Captured images are presented as one task-scoped collection with frame count, timestamps and keyframe/replay markers, preserving both useful and unusable observations.

Mission history remains visible when the operator submits another independent instruction. In Figure 4.8, the completed Navigate, Inspect and Report tasks, evidence folder, bounded perception conclusion and report-opening control remain available while a later navigation request is shown separately as current. This prevents results from being overwritten by the next command.

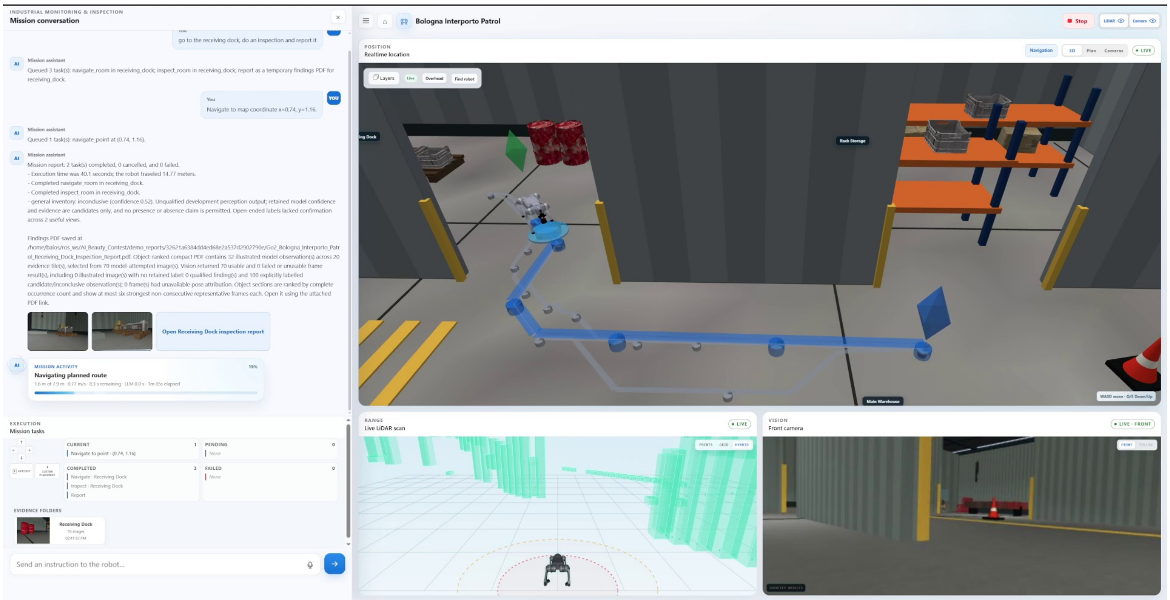

Figure 4.8: Persistent mission result and report hand-off. Completed tasks and zero failures remain in the task board; the conversation exposes execution details, the evidence qualification outcome, saved report path and a direct report-opening control.

The generated findings PDF turns the captured evidence into an operator-ready artifact. Object sections show observation count, matching-frame count, maximum and mean confidence, representative images and source identity. Candidate-only outputs are labelled explicitly and are not presented as qualified presence claims.

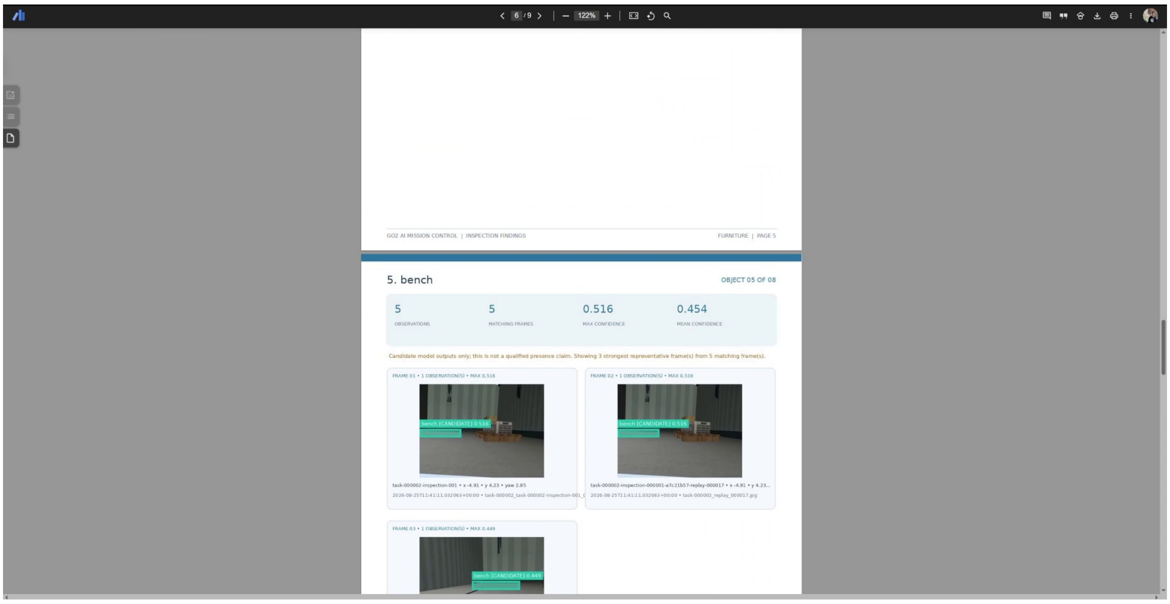

Figure 4.9: Evidence-based findings report. Each object section summarizes the supporting observations and confidence values and states when the result is only a candidate rather than a qualified presence claim.

4.6 Operate a Four-Robot Fleet

The same interface can supervise a fleet without creating four disconnected operator sessions. A single fleet objective is divided into non-duplicated room assignments. The fleet tab shows the shared state, while robot tabs expose each individual conversation and queue. Colour-coded routes, goals and robot markers make ownership visible in both the 2D plan and 3D views.

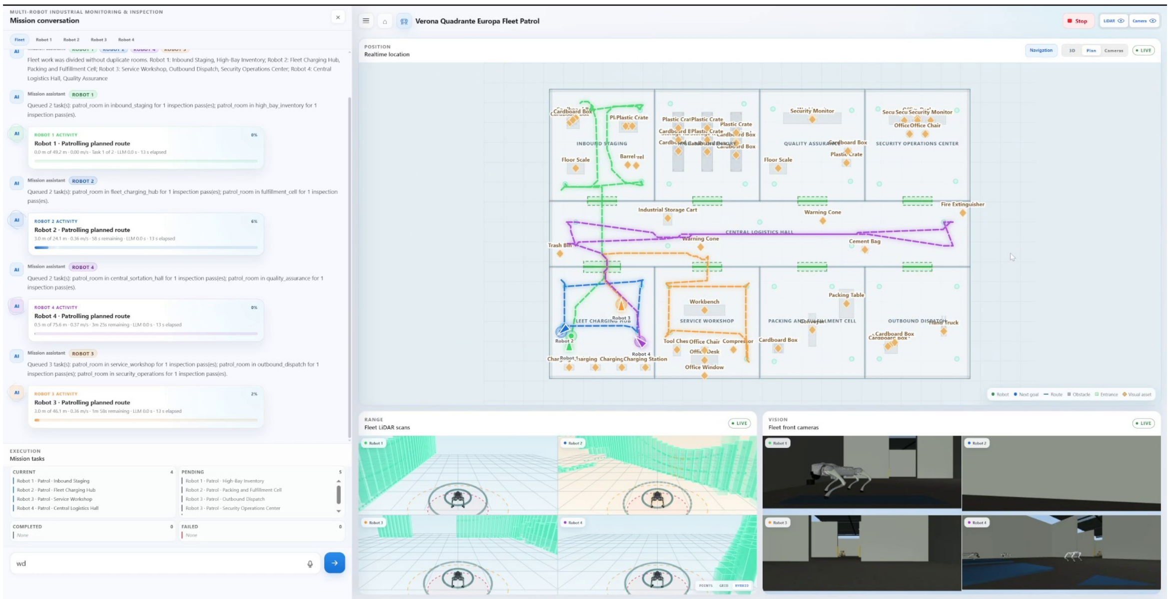

Figure 4.10: Fleet mission plan. Four task queues, colour-coded routes, assigned rooms, LiDAR panes and robot cameras are visible in one supervisory workspace.

Fleet safety is also visible to the operator. When routes interact, the coordinator can move one robot to a configured clearance position and hold it before resuming the admitted route. The activity stream reports this automatic route-clearing action, while the spatial view shows all robot positions and route ownership.

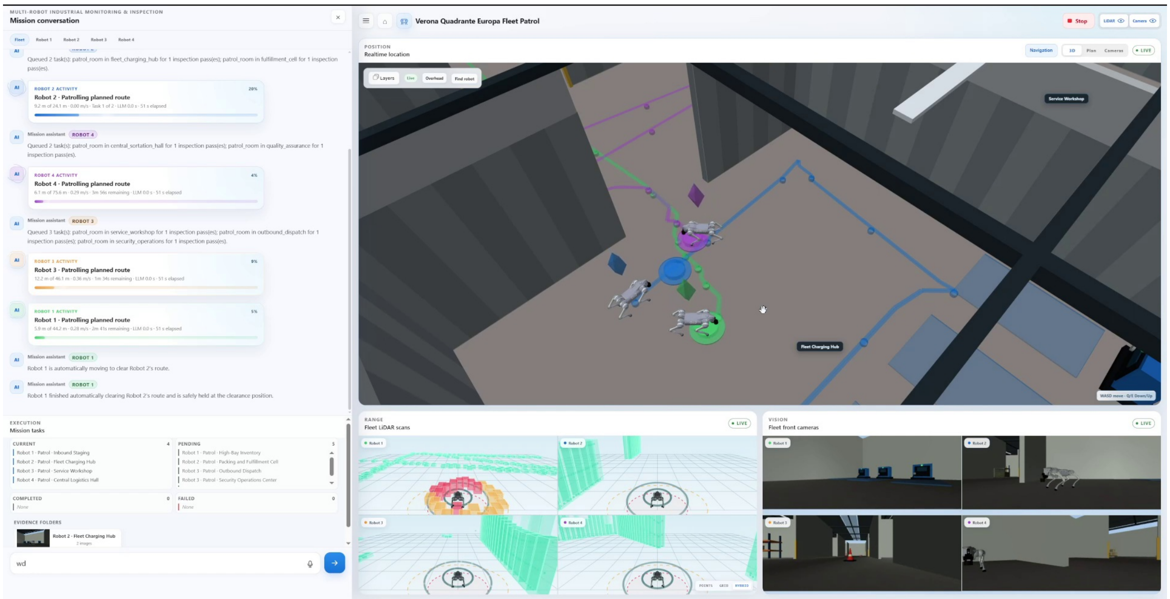

Figure 4.11: Automatic fleet coordination. The activity stream reports that Robot 1 clears Robot 2's route and reaches a safe hold position; the 3D view and per-robot sensor panes show the resulting spatial relationship.

The camera mode combines fixed environmental cameras with the four robot camera streams. Evidence folders stay accessible below the task board, allowing the operator to move directly from area-level awareness to the observations captured by a particular robot and room. Offline feeds are labelled rather than being replaced with stale imagery.

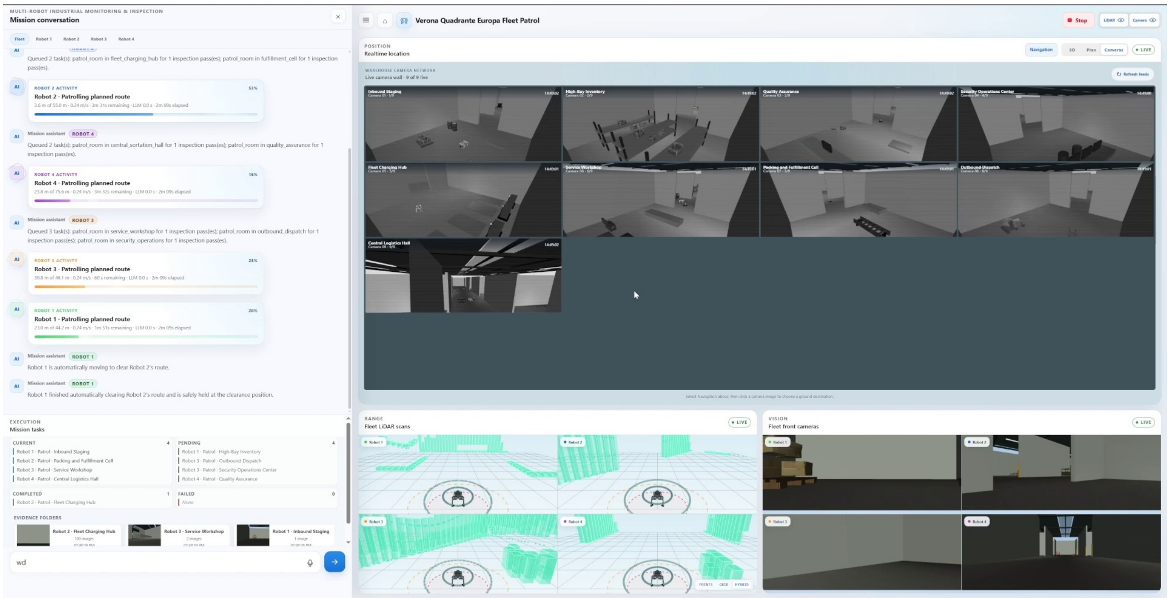

Figure 4.12: Fleet camera network and evidence overview. Fixed cameras, four robot cameras, task progress and room-scoped evidence folders are presented together; unavailable feeds are explicitly marked offline.

4.7 Select and Reuse Configured Deployments

The mission network separates reusable mission-control software from site-specific configuration. An operator first selects an available deployment from a map. The location card describes the mission, opens its workspace and shows previous deployment outcomes, including cancellations, so operational history is not hidden.

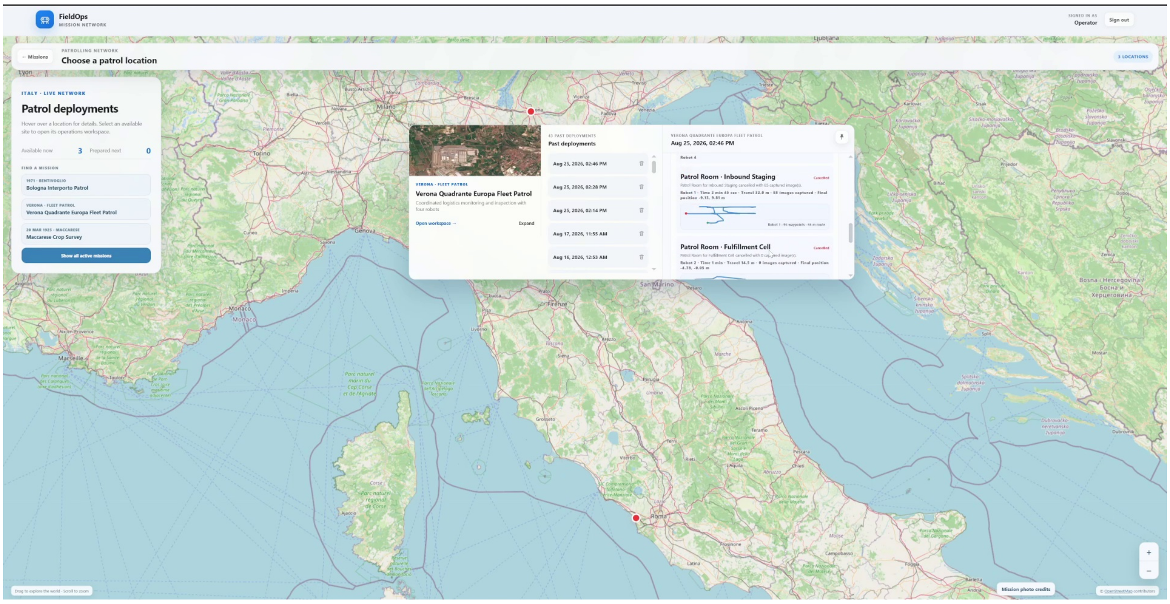

Figure 4.13: Deployment selection and history. The map presents available patrol locations, the selected site's purpose, a workspace entry point and prior mission outcomes with route summaries.

After selection, the same operator model is applied to the site's semantic map. The crop-survey configuration in Figure 4.14 includes a photogrammetric environment, named grid cells, automatically generated unsafe regions and a robot spawn location. These elements constrain later language requests without changing the core application.

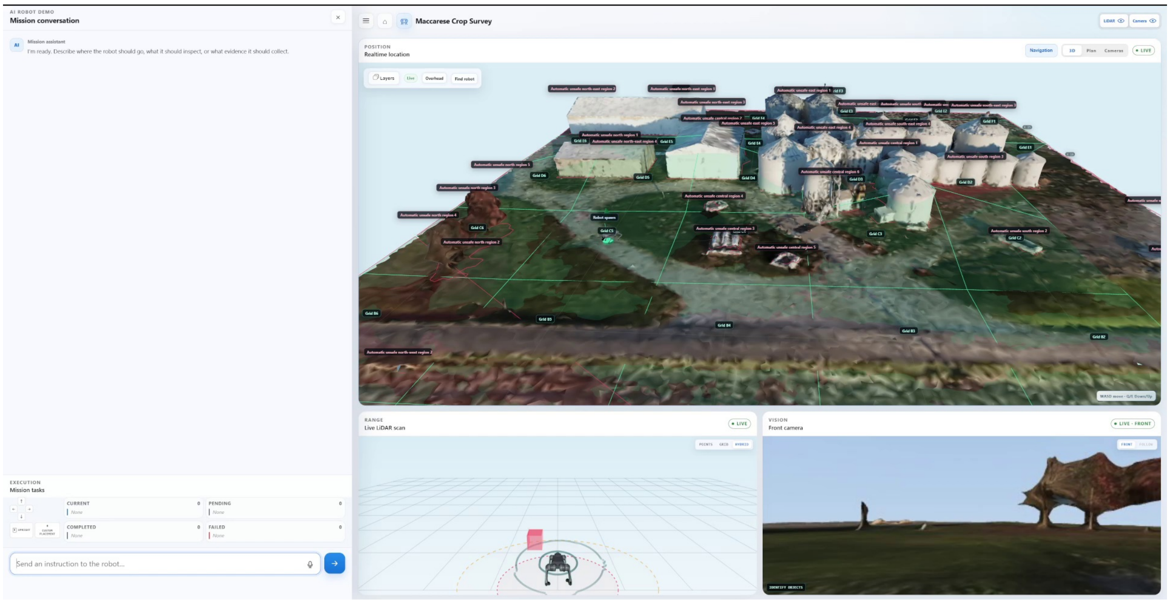

Figure 4.14: Configured crop-survey deployment. Site geometry, semantic grid, automatically derived unsafe regions, robot position, LiDAR and camera are loaded into the same mission workspace.

The real-site view in Figure 4.15 demonstrates the same navigation workflow on scanned terrain. A map-coordinate instruction is represented as a semantic task, the planned route is drawn on the reconstructed site, and the operator retains the same activity, task, LiDAR, camera and Stop controls used in the warehouse.

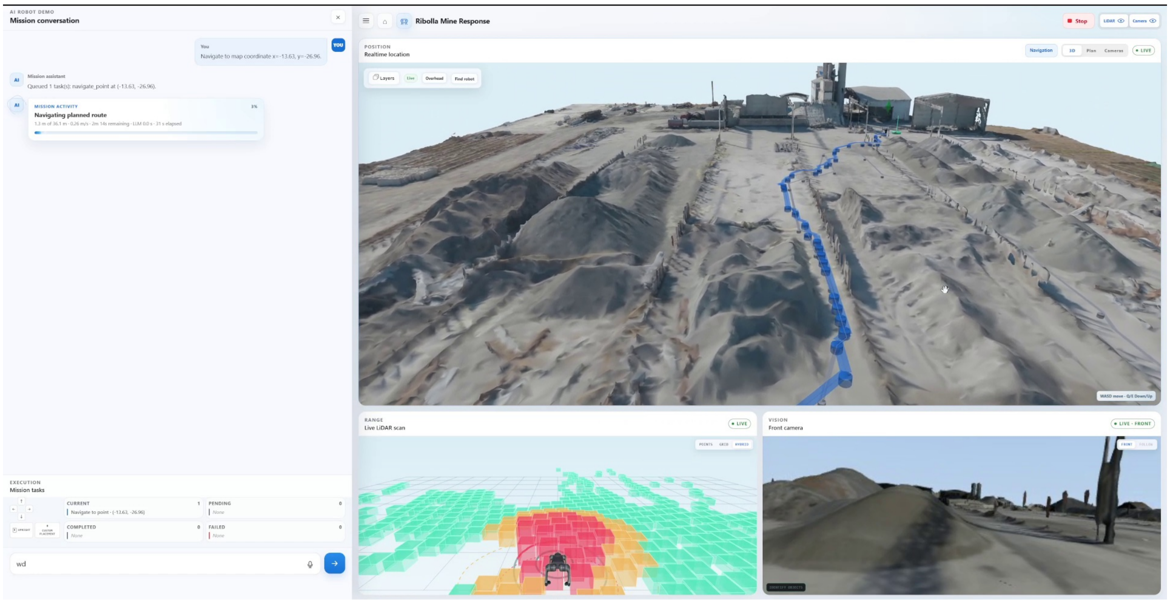

Figure 4.15: Navigation on a reconstructed real site. The route is displayed on the photogrammetric model while mission activity, task state, LiDAR, camera and operator authority remain unchanged.

4.8 End-to-End User Flow

The complete operator workflow is summarized in Table 4.1.

Table 4.1: How an operator uses the platform from operation selection to deliverable.

Step	Operator action	Platform response
1	Choose Patrolling or Rescuing.	Load the appropriate operational context and list of configured locations.
2	Select a configured deployment.	Load the site's semantic map, allowed areas, assets, sensors and mission history.
3	State the objective in natural language.	Resolve configured locations and assets, validate the proposal and expose the resulting semantic task queue.
4	Supervise execution.	Execute guarded ROS 2 actions while showing route, progress, LiDAR, camera and persistent Stop authority.
5	Review captured evidence.	Group observations by mission, robot, task and room while retaining keyframes and unusable-frame history.
6	Consume the result.	Publish terminal task state, bounded perception conclusions and a traceable findings PDF.
7	Scale supervision when required.	Allocate non-duplicated fleet work, coordinate route admission and combine robot and site-level sensing in one workspace.

The screenshots therefore document the complete usable loop:

> operation selection → deployment selection → natural-language objective → validated semantic tasks → guarded ROS 2 execution → evidence review → bounded findings report.

This chapter demonstrates the integrated platform workflow. Aggregate mission completion, human-intervention and perception rates remain separate measurements and are reported only in Chapter 5.

Chapter 5

KPIs and Results

5.1 Evaluation Protocol and Evidence Boundary

The measurements apply to MuJoCo simulation only. They were collected with ROS 2 Humble, Python 3.10.12 and MuJoCo 3.10.0 on Ubuntu 22.04 under WSL2, using an AMD Ryzen 7 3700X, an NVIDIA GeForce RTX 3070 with 8 GiB of memory and approximately 16 GiB of system memory. The single-robot Bologna profile used the guarded PGTT locomotion actor and a qualified local perception configuration. The four-robot Verona profile used a separate map and runtime configuration. These profiles are therefore kept as distinct evidence groups; the scenario matrix reports their results together, but does not treat unlike configurations as one interchangeable performance population.

The sealed protocol required five exact-wording trials for each of four scenarios. The three single-robot scenarios ran within one shared launch. Each four-robot trial used a fresh launch to preserve trial isolation. A nominal run was successful only when the expected semantic task set reached an authoritative terminal state, all required inspection viewpoints completed, all requested permanent images were created, any requested findings report was available, and no task remained unresolved. Failed, cancelled, timed-out or incomplete runs would have remained in the denominator.

Six KPI packages were produced: one containing the fifteen single-robot trials and five containing one fleet trial each. All six package manifests passed integrity validation with no hash mismatch and no uncommitted raw-log tail. Logical package completion is a stricter condition and remained false because the planned safety suite and formal screenshot set were not completed. This distinction is retained throughout the chapter.

5.2 KPI Definitions

For the nominal campaign, mission completion and unplanned human intervention were calculated as

$$\text{mission completion rate} = \frac{\text{successful nominal missions}}{\text{attempted nominal missions}}, \tag{5.1}$$

$$\text{human-intervention rate} = \frac{\text{nominal missions with at least one unplanned intervention}}{\text{attempted nominal missions}}. \tag{5.2}$$

Planned safety tests and unclassified runs are excluded from both denominators. Evidence coverage is the number of completed required viewpoints with accepted, identity-bound evidence divided by the number of required viewpoints. Report generation success is calculated only over missions that explicitly requested a findings PDF. Perception precision and recall use independently labelled qualification cases rather than live detections or semantic-map inventory as ground truth.

5.3 Nominal Mission Results

All twenty required nominal trials completed successfully. Each scenario therefore achieved five successes from five attempts, as shown in Table 5.1. The overall 20/20 result is the sealed scenario-matrix outcome; the per-scenario rows remain the appropriate basis for comparison because task content and runtime configuration differ.

Table 5.1: Results of the sealed nominal mission matrix. Duration is measured from browser instruction ingress to the authoritative terminal operation.

Scenario	Successful / attempted	Completion rate	Duration: min / median / max (s)
Receiving Dock barrel inspection with permanent images	5 / 5	100%	16.498 / 17.953 / 37.224
Maintenance Bay inspection for tool chest and air compressor	5 / 5	100%	72.848 / 77.332 / 83.594
Receiving Dock inspection with findings PDF	5 / 5	100%	146.635 / 150.708 / 152.161
Four-robot fleet warehouse inspection	5 / 5	100%	356.098 / 370.870 / 375.604
Sealed scenario matrix	**20 / 20**	**100% in every scenario**	Not pooled across unlike profiles

No nominal run failed, timed out or was cancelled. No run recorded an unplanned human intervention: the single-robot group measured $0/15$ and the fleet groups measured $0/5$. Correction and resubmission linkage was complete under the trusted campaign runner, so the human-intervention rate is 0 % within each configuration group. The run records also report zero measured navigation replans and zero recoveries. One observed internal replan transition occurred during each fleet run, but did not increment the authoritative completed-run replan counter or prevent mission completion.

5.4 Evidence and Reporting Results

The campaign required 125 inspection viewpoints: 45 across the fifteen single-robot trials and 80 across the five fleet trials. All 125 viewpoints completed and produced 125 accepted evidence records, giving 100 % evidence coverage in every configuration group. For the permanent-image scenario, all 15 expected images were created. For the report scenario, all five explicitly requested findings PDFs were generated and durably retained, giving a $5/5$, or 100 %, report-generation success rate.

The recorder observed every requested PDF as already durable when the related terminal mission state was committed. Under the implemented measurement boundary—terminal-state observation to durable report copy-and-hash commit—the measured availability latency was therefore 0 s for all five reports. This does not represent browser rendering time, network latency or the full report-generation duration; it shows only that no additional post-completion wait was observed at the recorder boundary.

Table 5.2 summarizes the principal measured indicators and their scope.

5.5 Perception Qualification

The final evaluation bundle contains a hash-verified perception qualification artifact matching the active single-robot perception configuration. Its scope is limited to the authorized `barrel` class and the equivalent `barrels` query. Each query was evaluated on five positive and five negative labelled cases across the required coverage conditions. The aggregate query-level confusion matrix was

$$TP = 10, \qquad TN = 10, \qquad FP = 0, \qquad FN = 0. \tag{5.3}$$

Precision, recall, negative correctness and classification coverage were all 100 %; there were no

Table 5.2: Principal KPI measurements and interpretation boundaries.

Indicator	Measured result	Interpretation
Nominal mission completion	5/5 in each scenario	Passed for all four exact-wording simulated scenarios.
Unplanned human intervention	0/15 single robot; 0/5 fleet	Complete measurement within the sealed nominal trials.
Inspection evidence coverage	125/125 viewpoints (100%)	Every required viewpoint had accepted, identity-bound evidence.
Permanent-image creation	15/15 images (100%)	Applies only to the scenario that explicitly requested retention.
Findings-report generation	5/5 PDFs (100%)	Applies only to missions that explicitly requested a findings PDF.
Navigation replans / recoveries	0 / 0 measured	Authoritative completed-run counters; fleet logs separately retained one observed replan transition per run.

candidate, unsupported or inconclusive outputs. Both query forms therefore passed the qualification criteria. These figures establish performance only for the qualified barrel scope in the frozen simulated scene and render configuration. They do not qualify the other objects named in the demonstration, other environments or physical-camera imagery. In addition, the individual KPI package records that an independently hashed calibration manifest was unavailable, so complete proof that calibration and final qualification cases were disjoint remains an evidence limitation.

5.6 Instruction Regression

The retained instruction-regression artifact passed all 37 cases, producing 100 % accuracy against a required minimum of 95 %. The mean response latency was 6.679 s and the maximum was 18.324 s. Of the 30 cases that expected the local model backend, 24 used it, giving 80 % local-backend coverage; the remaining valid outcomes used deterministic handling or clarification paths.

This is useful supporting evidence for correct task interpretation, but the final evaluation marks the regression artifact as not matching the active campaign configuration. It is therefore not used to elevate the campaign to a complete automatic-acceptance result.

Reproducing the campaign. The same twenty-trial sealed campaign can be run from a clean checkout by building and sourcing the workspace, creating external output directories and then starting the automatic campaign runner:

```
source /opt/ros/humble/setup.bash
source .venv/bin/activate
colcon build --packages-select go2_mujoco_sim go2_ai_demo --symlink-install
source install/setup.bash

export GO2_CAMPAIGN_DIR="$(dirname "$PWD")/go2_campaign"
export GO2_CAMPAIGN_RESULTS="$(dirname "$PWD")/go2_campaign_kpis"
export GO2_CAMPAIGN_PERCEPTION="$(dirname "$PWD")/go2_campaign_perception"

ros2 run go2_ai_demo kpi_campaign init \
  --workspace "$PWD" \
```

```
  --campaign-dir "$GO2_CAMPAIGN_DIR"

ros2 run go2_ai_demo kpi_campaign run-all \
  --manifest "$GO2_CAMPAIGN_DIR/campaign_manifest.json" \
  --journal "$GO2_CAMPAIGN_DIR/campaign_journal.jsonl" \
  --results-root "$GO2_CAMPAIGN_RESULTS" \
  --perception-root "$GO2_CAMPAIGN_PERCEPTION"
```

The runner performs the perception qualification, instruction regression and all fifteen single-robot and five fleet trials, while retaining the resulting KPI evidence below the external results directory. For an ordinary interactive mission rather than the complete sealed campaign, KPI recording can be enabled and directed to an external location as follows:

```
export GO2_KPI_RESULTS="$(dirname "$PWD")/go2_kpi_results"

ros2 launch go2_ai_demo mission_portal.launch.py \
  kpi_capture_enabled:=true \
  kpi_results_root:="$GO2_KPI_RESULTS"
```

Each launched runtime creates a new timestamped evidence package in the selected results location; existing packages are not overwritten.

Chapter 6

Business Case

6.1 Scope and Commercial Recommendation

The current application is an integrated simulation prototype rather than a production-ready physical inspection system. The initial commercial offer should therefore remain close to the demonstrated level of maturity.

The proposed first offer is a **paid simulation-first inspection pilot for one camera-instrumented site**. Structured indoor facilities are the preferred initial setting because fixed cameras, controlled access, repeatable lighting and bounded operating areas simplify reconstruction and supervision. The same platform can also support bounded outdoor use cases when camera or survey coverage, localization, connectivity, environmental conditions and site-specific safety constraints are adequate. The warehouse in this report is therefore a representative example, not the product boundary.

During the pilot, the customer's inspection workflow is represented in a configured simulation, translated into bounded robotic missions and evaluated using agreed technical and operational KPIs. The pilot produces a validated mission workflow, an operator demonstration, inspection evidence and a plan for subsequent physical integration.

> **Commercial recommendation:** offer a bounded, simulation-first inspection pilot for a camera-instrumented indoor or outdoor site as an engineering and validation service, followed by site onboarding and an annual mission-control software licence when physical deployment becomes technically and commercially justified.

This approach generates initial service revenue while allowing the reusable software platform to develop into a more scalable licence-based product. It also avoids presenting simulation results as evidence of physical-robot readiness.

6.2 Initial Customer and Operational Problem

The initial target customer is a facility, infrastructure or site operator that performs recurring visual inspections and is already evaluating mobile robotics or working with a robotics systems integrator. Suitable indoor examples include warehouses, industrial buildings, utility rooms and other structured facilities. Bounded outdoor examples include logistics yards, agricultural areas, industrial perimeters and infrastructure sites.

A suitable first customer has:

- one camera-instrumented or surveyable site with clearly defined inspection areas;
- recurring and documentable inspection workflows;
- an operations or safety manager responsible for the process;
- an automation team or integration partner;
- a willingness to compare the proposed workflow with the current manual procedure.

The first implementation example is a recurring warehouse inspection covering areas such as receiving, storage and shipping. The robot is requested to visit configured viewpoints, collect visual evidence and produce a findings report when explicitly requested. The commercial product, however, is the reusable inspection workflow and mission-control layer rather than a warehouse-only application.

The expected value is not based on replacing the complete human inspection process. It is based on reducing routine operator effort, improving repeatable coverage and simplifying the association of evidence with a mission, location and time.

6.3 Proposed Commercial Offer

The pilot is intentionally limited to one site and a small number of inspection workflows. This limits delivery risk and makes the result measurable.

Table 6.1 defines the proposed pilot boundary.

Table 6.1: Proposed scope of the initial commercial pilot.

Element	Proposed scope
Duration	Eight to twelve weeks.
Site	One camera-instrumented indoor facility or one bounded outdoor area with adequate capture and localization coverage.
Workflows	Up to three inspection, patrolling or monitoring missions.
Configuration	Semantic areas, inspection viewpoints, authorized visual targets, reconstruction inputs and reporting rules.
Demonstration	Natural-language mission creation, supervised execution, evidence review and findings-report generation.
Validation	Agreed mission-completion, intervention and perception KPIs.
Training	Two operator training and review sessions.
Exclusions	Physical-robot certification, production safety approval, enterprise integration and hardware procurement.

The pilot should conclude with a decision on whether to stop, repeat the validation, proceed to physical integration or prepare a limited production deployment.

6.4 Cost and Pricing Assumptions

No customer quotations or measured delivery costs are currently available. Consequently, the following values are **planning assumptions** used to construct an initial financial case. They are not presented as market-validated prices.

Eurostat estimated average 2024 hourly labour costs at EUR 33.5 in the EU and EUR 37.3 in the euro area, with country values ranging from EUR 10.6 to EUR 55.2 [1]. The model therefore uses a fully loaded specialist project rate of **EUR 50 per hour**. This is a budgeting assumption above the EU and euro-area averages to allow for specialist engineering and employer overhead; it is not a quoted market rate.

Hardware purchase is excluded because the first pilot is simulation-first. A physical robot, additional sensors and on-site compute would require a separate quotation.

Table 6.2 itemizes the resulting direct-cost estimate.

Table 6.2: Estimated direct cost of one simulation-first pilot.

Cost item	Estimated cost
Workflow discovery and requirements, 40 hours	EUR 2.000
Site and mission configuration, 100 hours	EUR 5.000
Testing and KPI measurement, 80 hours	EUR 4.000
Documentation and operator training, 24 hours	EUR 1.200
Project management and pilot support, 36 hours	EUR 1.800
Compute, travel and pilot materials	EUR 2.000
Contingency, 15% of direct cost	EUR 2.400
Estimated pilot delivery cost	**EUR 18.400**

A proposed pilot price of **EUR 25.000** produces an estimated direct contribution of:

$$\text{EUR } 25.000 - \text{EUR } 18.400 = \text{EUR } 6.600. \tag{6.1}$$

The corresponding contribution margin is approximately 26%. This margin is intentionally moderate because early pilots contain significant customer-specific engineering.

Public robotics benchmarks are useful only as directional checks. Knightscope reported annual autonomous-security-robot subscriptions of US$1–11 per operating hour in a 2025 investor presentation [2]. At continuous utilization this corresponds to approximately US$8,760–96,360 per year, but that offer includes robot hardware and associated services and is therefore broader than the software-and-support offer proposed here. The comparison supports a recurring service model; it does not validate the proposed price.

If a pilot progresses to a limited production deployment, the proposed commercial structure is shown in Table 6.3.

The annual recurring package therefore produces EUR 16.000 of revenue and an estimated EUR 11.000 contribution per site before fixed development costs, general administration and tax. Robot hardware, maintenance, insurance and physical integration are excluded and must be supplied or priced separately.

Table 6.3: Initial pricing and direct-cost assumptions.

Commercial item	Revenue	Direct cost	Contribution
Simulation-first inspection pilot	EUR 25.000	EUR 18.400	EUR 6.600
Production-site onboarding	EUR 8.000	EUR 4.000	EUR 4.000
Annual software licence	EUR 12.000	EUR 2.000	EUR 10.000
Annual support package	EUR 4.000	EUR 3.000	EUR 1.000

6.5 Expected Revenue Scenarios

Table 6.4 presents three illustrative first-year scenarios. A conversion represents a pilot customer that purchases site onboarding and one annual software and support package. The calculation assumes that converted customers are invoiced early enough for the first annual package to be recognized within the same year.

Table 6.4: Illustrative first-year revenue scenarios.

Measure	Conservative	Base	Upside
Paid pilots	1	3	6
Production conversions	0	1	3
Expected revenue	EUR 25.000	EUR 99.000	EUR 222.000
Estimated direct cost	EUR 18.400	EUR 64.200	EUR 137.400
Estimated contribution	EUR 6.600	EUR 34.800	EUR 84.600
Contribution margin	26%	35%	38%
Year-end recurring revenue	EUR 0	EUR 16.000	EUR 48.000

Comparable-company revenue shows that recurring robotics and spatial-data services can generate material revenue, but it does not establish demand for this application. Knightscope reported 2025 total revenue of US$11.335 million, including US$7.968 million of service revenue and US$4.402 million of autonomous-security-robot revenue; it also reported service cost of revenue of US$12.324 million [3]. Matterport, whose platform combines 3D capture, digital twins, subscriptions and capture services, reported 2024 subscription revenue of US$99.590 million, services revenue of US$41.264 million and total revenue of US$169.699 million [4].

These benchmarks demonstrate existing commercial models for recurring robotic services and spatial-data platforms. They also show why revenue alone is not a profitability proxy. The scenarios in Table 6.4 remain arithmetic consequences of the proposed prices and assumed pilot counts; they are not sales forecasts or estimates derived from the comparable companies. Customer interviews and priced pilot discussions are required before conversion rates and achievable prices can be considered reliable.

6.6 Illustrative Customer Value

The customer case should be evaluated against the existing inspection process. For illustration, consider a camera-instrumented facility performing four inspection and documentation rounds per day over 250 working days. If each round requires 1.25 hours at a fully loaded labour cost of EUR 35 per hour, the annual baseline cost is:

$$4 \times 250 \times 1.25 \times \text{EUR } 35 = \text{EUR } 43.750. \tag{6.2}$$

If supervised robotic execution reduces active human time by 50%, the potential annual labour saving is EUR 21.875. After an annual software and support cost of EUR 16.000, the remaining measurable annual benefit is EUR 5.875.

Under these assumptions, the EUR 8.000 production-site onboarding fee has a simple payback period of approximately 16 months:

$$\frac{\text{EUR } 8.000}{\text{EUR } 5.875/12} \approx 16.3 \text{ months.} \tag{6.3}$$

This calculation excludes robot purchase, leasing and physical integration. It is therefore most relevant to a customer that already owns a compatible robot or obtains it through an integration partner. It also excludes possible benefits from increased coverage, improved evidence quality, reduced exposure to hazardous areas and avoided downtime because these benefits have not yet been measured.

6.7 Scalability

Early deployments will require substantial engineering effort. The commercial model becomes more scalable only if customer-specific work is progressively replaced by reusable software and standard procedures.

The main scaling mechanisms are:

- configuration-driven semantic maps and inspection viewpoints;
- reusable mission and skill contracts;
- standard indoor and outdoor inspection mission templates;
- repository-local language and perception models;
- repeatable site-onboarding and acceptance procedures;
- annual site licences and standardized support packages;
- delivery through robotics and systems-integration partners.

The principal scaling risk is excessive customization. If every customer requires a different robot interface, map format, perception model and reporting workflow, the business remains an engineering service with limited margins. A pilot should therefore measure configuration and support effort as carefully as mission performance.

6.8 Risks and Decision Gates

Table 6.5 connects each principal commercial or deployment risk to the evidence and decision gate required before expansion.

6.9 Business-Case Conclusion

The recommended first commercial step is a EUR 25.000 simulation-first inspection pilot for one camera-instrumented site. Under the stated assumptions, the pilot costs approximately EUR 18.400 to deliver and produces a direct contribution of EUR 6.600. A successful pilot can progress to EUR 8.000 of production-site onboarding and EUR 16.000 of annual recurring software and support revenue.

The base first-year scenario consists of three paid pilots and one production conversion, producing an estimated EUR 99.000 in revenue and EUR 34.800 in direct contribution. These figures are planning assumptions rather than validated forecasts.

Table 6.5: Principal business risks and corresponding decision gates.

Risk	Evidence required	Decision gate
Insufficient customer need	Interviews and measurement of the current inspection workflow.	Do not start a pilot without a frequent and measurable problem.
Low willingness to pay	A priced proposal reviewed by a qualified customer.	Proceed only with a paid pilot or an explicitly funded validation.
Excessive delivery effort	Actual hours spent on mapping, configuration, testing and support.	Continue only if later sites can reuse a substantial part of the work.
Physical integration risk	Robot interface, localization, sensor and emergency-stop validation.	Do not claim production readiness before physical acceptance testing.
Outdoor operating conditions	Weather, lighting, terrain, communications and capture coverage.	Approve outdoor use only after site-specific environmental and safety validation.
Unqualified perception	Customer-site ground truth and measured false-positive and false-negative results.	Keep findings advisory until the required class passes qualification.
Licensing and security gaps	Review of software, models, assets, data handling and deployment security.	Resolve material issues before external commercial deployment.

The business case is attractive only if three conditions are demonstrated: customers are willing to pay for the pilot, customer-specific delivery effort decreases across sites, and physical operation can meet the agreed safety, mission and perception requirements. The next commercial action is therefore to present the bounded pilot to suitable facility operators or integration partners and replace the assumptions in this chapter with measured quotations, delivery hours and customer baseline data.

References

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